PHYSICAL CONSTANTS USED IN TRANSMISSION LINE WORK    file: constant.mcd

Electric permittivity of free space (metric)    $E0\_meters := 8.854 \cdot 10^{-12}$    F/m

Recalculate in in.    $E0\_inches := E0\_meters \cdot .0254$

Display calculated value    $E0\_inches = 2.249 \times 10^{-13}$

Magnetic permeability of free space (metric)    $U0\_meters := 4 \cdot \pi \cdot 10^{-7}$    H/m

Recalculate in in.    $U0\_inches := U0\_meters \cdot .0254$

Display calculated value    $U0\_inches = 3.192 \times 10^{-8}$

We often need this number    $\frac{U0\_inches}{2 \cdot \pi} = 5.08 \times 10^{-9}$

Speed of light (metric)    $C\_meters := 2.998 \cdot 10^{8}$    m/s

Recalculated in in.    $C\_inches := \frac{C\_meters}{.0254}$

Display calculated value    $C\_inches = 1.18 \times 10^{10}$

Propagation delay at light speed (ps/in.)    $\frac{10^{12}}{C\_inches} = 84.723$